WORKED SOLUTION

STEP III 2025, Question 8

Pure • Trigonometric Sums via Complex Numbers

STEP III 2025, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

The identity in (i) is a recurrence: it writes $z^{m+1} - z^{-(m+1)}$ in terms of the same kind of expression two steps lower, so applying it repeatedly makes a sum telescope out. With $z = \mathrm{e}^{\mathrm{i}\theta}$ every $z^k \pm z^{-k}$ becomes $2\cos k\theta$ or $2\mathrm{i}\sin k\theta$, turning the algebra into a trigonometric identity. The numerical parts then come from choosing θ so that $\sin 2n\theta$ equals $\sin\theta$ (or $\cos\theta$) and cancels.

Solution

Part (i)

Expanding the product,

$$\left(z - \frac{1}{z}\right)\left(z^m + \frac{1}{z^m}\right) = z^{m+1} + z^{1-m} - z^{m-1} - z^{-(m+1)}.$$

Adding $z^{m-1} - z^{1-m}$ cancels the two middle terms and leaves $z^{m+1} - z^{-(m+1)}$, which is the required identity.

Induction. For $n = 1$ the claim is $z^2 - z^{-2} = (z - z^{-1})(z + z^{-1})$, the difference of two squares, so it holds. Suppose it holds for some $n \geq 1$. Using the identity with $m = 2n + 1$ (so $m + 1 = 2n + 2$ and $m - 1 = 2n$),

$$\begin{aligned} z^{2n+2} - \frac{1}{z^{2n+2}} &= \left(z - \frac{1}{z}\right)\left(z^{2n+1} + \frac{1}{z^{2n+1}}\right) + \left(z^{2n} - \frac{1}{z^{2n}}\right) \\ &= \left(z - \frac{1}{z}\right)\left(z^{2n+1} + \frac{1}{z^{2n+1}}\right) + \left(z - \frac{1}{z}\right)\sum_{r=1}^{n}\left(z^{2r-1} + \frac{1}{z^{2r-1}}\right) \\ &= \left(z - \frac{1}{z}\right)\sum_{r=1}^{n+1}\left(z^{2r-1} + \frac{1}{z^{2r-1}}\right), \end{aligned}$$

by the inductive hypothesis, since the extra term is the $r = n + 1$ term. So the result holds for $n + 1$, and by induction for all $n \geq 1$.

The version with $z+\frac{1}{z}$. In the same way,

$$\left(z+\frac{1}{z}\right)\left(z^m-\frac{1}{z^m}\right)=z^{m+1}-z^{1-m}+z^{m-1}-z^{-(m+1)},$$

so this time we must *subtract* $z^{m-1}-z^{1-m}$:

$$z^{m+1}-\frac{1}{z^{m+1}}=\left(z+\frac{1}{z}\right)\left(z^m-\frac{1}{z^m}\right)-\left(z^{m-1}-\frac{1}{z^{m-1}}\right).$$

The minus sign makes the signs alternate as we step down, and we claim

$$z^{2n}-\frac{1}{z^{2n}}=\left(z+\frac{1}{z}\right)\sum_{r=1}^{n}(-1)^{n-r}\left(z^{2r-1}-\frac{1}{z^{2r-1}}\right).$$

For $n=1$ this is again $(z+z^{-1})(z-z^{-1})$. If it holds for n, the identity with $m=2n+1$ gives

$$z^{2n+2}-\frac{1}{z^{2n+2}}=\left(z+\frac{1}{z}\right)\left[\left(z^{2n+1}-\frac{1}{z^{2n+1}}\right)-\sum_{r=1}^{n}(-1)^{n-r}\left(z^{2r-1}-\frac{1}{z^{2r-1}}\right)\right],$$

and since $-(-1)^{n-r}=(-1)^{n+1-r}$, while the first term is the $r=n+1$ term with sign $(-1)^0=1$, the bracket is exactly $\sum_{r=1}^{n+1}(-1)^{n+1-r}(z^{2r-1}-z^{-(2r-1)})$. This completes the induction. Written out,

$$z^{2n}-\frac{1}{z^{2n}}=\left(z+\frac{1}{z}\right)\left[\left(z^{2n-1}-\frac{1}{z^{2n-1}}\right)-\left(z^{2n-3}-\frac{1}{z^{2n-3}}\right)+\cdots\pm\left(z-\frac{1}{z}\right)\right].$$

Part (ii)

(a) With $z=\mathrm{e}^{\mathrm{i}\theta}$ we have $z^k+z^{-k}=2\cos k\theta$ and $z^k-z^{-k}=2\mathrm{i}\sin k\theta$. The first result of (i) becomes

$$2\mathrm{i}\sin 2n\theta=2\mathrm{i}\sin\theta\sum_{r=1}^{n}2\cos(2r-1)\theta,$$

and dividing by 2i gives $\sin 2n\theta=2\sin\theta\sum_{r=1}^{n}\cos(2r-1)\theta$, as required.

(b) With $n=2$: $\sin 4\theta=2\sin\theta(\cos\theta+\cos 3\theta)$. We want $\sin 4\theta$ to equal $\sin\theta$, and $\theta=\frac{1}{5}\pi$ does this, because $\sin\frac{4}{5}\pi=\sin(\pi-\frac{1}{5}\pi)=\sin\frac{1}{5}\pi$. Since $\sin\frac{1}{5}\pi\neq 0$ we may divide by it:

$$1=2(\cos\tfrac{1}{5}\pi+\cos\tfrac{3}{5}\pi).$$

Now $\cos\frac{3}{5}\pi=-\cos(\pi-\frac{3}{5}\pi)=-\cos\frac{2}{5}\pi$, so $\cos\frac{1}{5}\pi-\cos\frac{2}{5}\pi=\frac{1}{2}$, that is, $\cos\frac{2}{5}\pi=\cos\frac{1}{5}\pi-\frac{1}{2}$.

(c) With $n=7$ and $\theta=\frac{1}{15}\pi$, again $\sin 14\theta=\sin(\pi-\theta)=\sin\theta\neq 0$, so

$$\sum_{r=1}^{7}\cos\frac{(2r-1)\pi}{15}=\frac{1}{2}.$$

Ask yourself The seven angles here are odd multiples of $\frac{1}{15}\pi$, but the target has even ones. How do we get from one to the other?

Use $\cos(\pi-\alpha) = -\cos\alpha$ on the angles beyond $\frac{1}{2}\pi$, and note $\cos\frac{5}{15}\pi = \cos\frac{1}{3}\pi = \frac{1}{2}$, and $\cos\frac{3}{15}\pi = \cos\frac{1}{5}\pi$:

$$\cos\tfrac{9}{15}\pi = -\cos\tfrac{2}{5}\pi, \qquad \cos\tfrac{11}{15}\pi = -\cos\tfrac{4}{15}\pi, \qquad \cos\tfrac{13}{15}\pi = -\cos\tfrac{2}{15}\pi.$$

The sum becomes

$$\cos\tfrac{1}{15}\pi + \cos\tfrac{7}{15}\pi - \cos\tfrac{2}{15}\pi - \cos\tfrac{4}{15}\pi + \tfrac{1}{2} + (\cos\tfrac{1}{5}\pi - \cos\tfrac{2}{5}\pi) = \tfrac{1}{2}.$$

By (b) the bracket is $\frac{1}{2}$, so

$$\cos\tfrac{2}{15}\pi + \cos\tfrac{4}{15}\pi - \cos\tfrac{7}{15}\pi - \cos\tfrac{1}{15}\pi = \tfrac{1}{2}.$$

Finally $-\cos\frac{7}{15}\pi = \cos(\pi - \frac{7}{15}\pi) = \cos\frac{8}{15}\pi$ and $-\cos\frac{1}{15}\pi = \cos(\pi + \frac{1}{15}\pi) = \cos\frac{16}{15}\pi$, giving

$$\cos\tfrac{2}{15}\pi + \cos\tfrac{4}{15}\pi + \cos\tfrac{8}{15}\pi + \cos\tfrac{16}{15}\pi = \tfrac{1}{2},$$

as required.

Part (iii)

Ask yourself The target is an alternating sum of sines of odd multiples of θ. Which result from (i) produces exactly that?

The $z + \frac{1}{z}$ version. With $z = \mathrm{e}^{\mathrm{i}\theta}$ it becomes

$$2\mathrm{i}\sin 2n\theta = 2\cos\theta \sum_{r=1}^{n} (-1)^{n-r}\, 2\mathrm{i}\sin(2r-1)\theta,$$

so

$$\sin 2n\theta = 2\cos\theta \sum_{r=1}^{n} (-1)^{n-r} \sin(2r-1)\theta.$$

Take $n = 3$:

$$\sin 6\theta = 2\cos\theta\,(\sin 5\theta - \sin 3\theta + \sin\theta).$$

Now we want $\sin 6\theta = \cos\theta$, and $\theta = \frac{1}{14}\pi$ works: $\sin\frac{6}{14}\pi = \cos(\frac{1}{2}\pi - \frac{6}{14}\pi) = \cos\frac{1}{14}\pi$, which is non-zero. Dividing by it,

$$\sin\tfrac{1}{14}\pi - \sin\tfrac{3}{14}\pi + \sin\tfrac{5}{14}\pi = \tfrac{1}{2}.$$

Checks

- Test the identities at a simple value: $z = 2$, $n = 2$ gives $16 - \frac{1}{16} = \frac{255}{16}$, while $(2 - \frac{1}{2})(2 + \frac{1}{2} + 8 + \frac{1}{8}) = \frac{3}{2} \cdot \frac{85}{8} = \frac{255}{16}$ and $(2 + \frac{1}{2})(8 - \frac{1}{8} - 2 + \frac{1}{2}) = \frac{5}{2} \cdot \frac{51}{8} = \frac{255}{16}$.

- Numerically, $\cos\frac{2}{5}\pi = 0.309017\ldots$ and $\cos\frac{1}{5}\pi - \frac{1}{2} = 0.809017\ldots - 0.5$ agree; this is the familiar $\cos\frac{1}{5}\pi = \frac{1+\sqrt{5}}{4}$, $\cos\frac{2}{5}\pi = \frac{\sqrt{5}-1}{4}$.
- The sums in (ii)(c) and (iii) both evaluate to 0.500000 on a calculator.

Common slips

- Getting the sign wrong in the $z + \frac{1}{z}$ recurrence: the lower term is subtracted, so the sum alternates and must end with the $+$ sign on the top term $z^{2n-1} - z^{-(2n-1)}$.
- Dividing by $\sin\theta$ or $\cos\theta$ without noting it is non-zero at the chosen angle.
- In (ii)(c), forgetting that $\cos\frac{5}{15}\pi = \frac{1}{2}$ and that (b) is needed to clear the remaining pair of terms.

Beyond STEP

The expressions $z^k + z^{-k}$ with $z = \mathrm{e}^{\mathrm{i}\theta}$ are $2\cos k\theta$, and recurrences like the one in (i) are exactly those satisfied by Chebyshev polynomials, which turn $\cos k\theta$ and $\sin k\theta / \sin\theta$ into polynomials in $\cos\theta$. The neat sums in (ii)(c) and (iii) come from roots of unity: the angles $\frac{2}{15}\pi$, $\frac{4}{15}\pi$, $\frac{8}{15}\pi$, $\frac{16}{15}\pi$ are generated by repeated doubling, a first glimpse of Gauss's periods and Galois theory. Needham's *Visual Complex Analysis* is a beautiful place to see this geometry.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.